

Station locations and prediction grid

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```
rm(list = ls())
```

```
library(SpatialTools)
```

```
## # This research was partially supported under NSF Grant ATM-0534173
```

```
library(fields)
```

```
## Loading required package: spam
```

```
## Spam version 2.11-1 (2025-01-20) is loaded.  
## Type 'help( Spam)' or 'demo( spam)' for a short introduction  
## and overview of this package.  
## Help for individual functions is also obtained by adding the  
## suffix '.spam' to the function name, e.g. 'help( chol.spam)'.
```

```
##  
## Attaching package: 'spam'
```

```
## The following object is masked from 'package:SpatialTools':  
##  
##      rmvnorm
```

```
## The following objects are masked from 'package:base':  
##  
##      backsolve, forwardsolve
```

```
## Loading required package: viridisLite
```

```
##  
## Try help(fields) to get started.
```

```
mydir <- "/home/arnab/Documents/delhi_aqi_data"  
setwd(mydir)
```

```
set.seed(1010)
```

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union

load("delhi_aqi_data.Rdata")

pm2.5_avg <- sapply(1:40, function(i){mean(DL.data[[i]]$pm2.5, na.rm = T)})

#-----

library(ggmap)

## Loading required package: ggplot2

## i Google's Terms of Service: <https://mapsplatform.google.com>
##   Stadia Maps' Terms of Service: <https://stadiamaps.com/terms-of-service>
##   OpenStreetMap's Tile Usage Policy: <https://operations.osmfoundation.org/policies/tiles>
## i Please cite ggmap if you use it! Use 'citation("ggmap")' for details.

library(viridis)

map.points <- function (lat, lon, data, mainTitle = NULL, legendTitle = NULL,
                        xlim = NULL, ylim = NULL, zlim = NULL){

  # Set limits for x, y, z if not specified as parameters
  if(is.null(xlim)){xlim <- range(lon, na.rm = TRUE)}
  if(is.null(ylim)){ylim <- range(lat, na.rm = TRUE)}
  if(is.null(zlim)){zlim <- range(data, na.rm = TRUE)}

  ukbox <- make_bbox(lon = c(min(lon), max(lon)), lat = c(max(lat), min(lat)), f = .1)
  uk <- get_map(location = ukbox, zoom = 10, maptype = "terrain", source = "google")
  dataframe <- data.frame(x = lon, y = lat, data = data)

  p <- ggmap(uk)
  p <- p + geom_point(dataframe, mapping = aes(x = x, y = y, color = data),
                     size = 2, alpha = 1) + #
    labs(title=mainTitle, x="Longitude", y="Latitude") +
    theme(axis.text=element_text(size=18),
          axis.title=element_text(size=18),
          plot.title = element_text(size=20, hjust = 0.5),
          legend.text = element_text(size = 18),
          legend.title = element_text(size = 18),
          legend.key.size = unit(0.3, "in")) +
    scale_color_viridis(name = legendTitle, limits = zlim) +
    xlim(xlim) + ylim(ylim)

  baseData <- map_data("world")
  p <- p + geom_path(data=baseData, aes(x=long, y=lat, group=group),
                    colour="black")
}
```

```

return(p)}

p <- map.points(lat = DL.details$latitude,
                lon = DL.details$longitude,
                data = pm2.5_avg,
                mainTitle = "PM2.5 conc (ug/m3)")

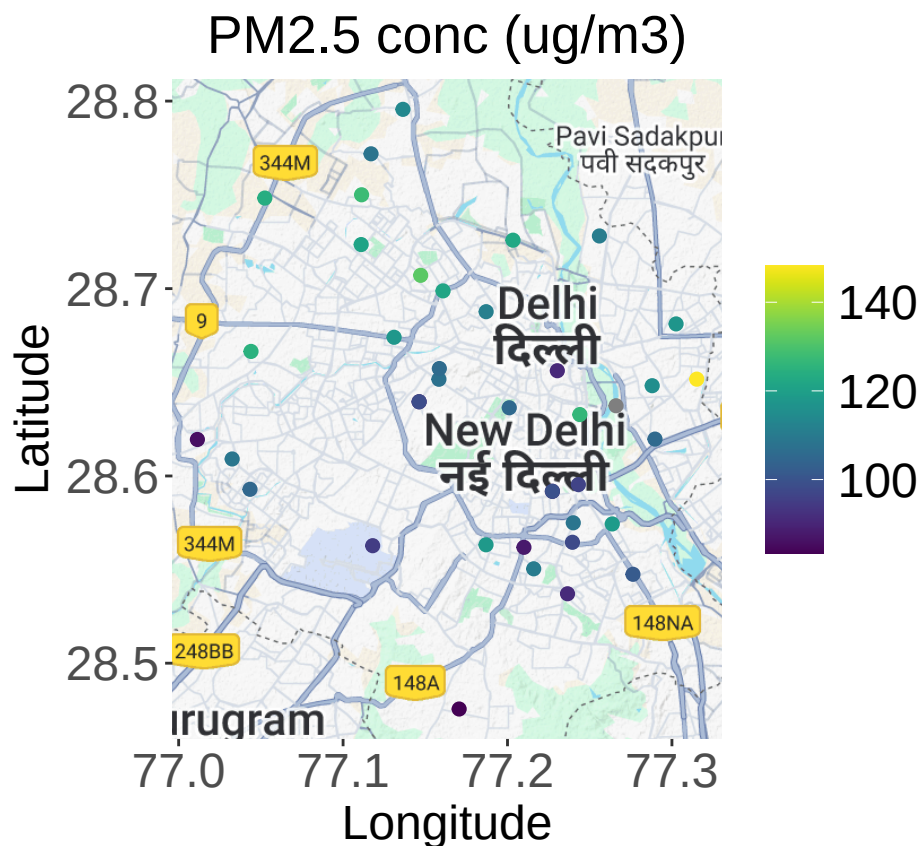
## ! Bounding box given to Google - spatial extent only approximate.
## i <https://maps.googleapis.com/maps/api/staticmap?center=28.63555,77.16325&zoom=10&size=640x640&scal
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.

print(p)

## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_rect()').

## Warning: Removed 99338 rows containing missing values or values outside the scale range
## ('geom_path()').

```



Imagine we just use inverse distance weighting for prediction.

```

library(fields)

distmat <- rdist(cbind(DL.details$longitude, DL.details$latitude), locs.pred)

remove.case <- which(is.na(pm2.5_avg))

distmat <- distmat[-(remove.case), ]

weights <- (1 / distmat)^0.25 / colSums((1 / distmat)^0.25)

pm2.5.pred <- as.vector(t(weights) %*% pm2.5_avg[-(remove.case)])

map.heatmap <- function (lat, lon, data, mainTitle = NULL, legendTitle = NULL,
                          xlim = NULL, ylim = NULL, zlim = NULL){

  # Set limits for x, y, z if not specified as parameters
  if(is.null(xlim)){xlim <- range(lon, na.rm = TRUE)}
  if(is.null(ylim)){ylim <- range(lat, na.rm = TRUE)}
  if(is.null(zlim)){zlim <- range(data, na.rm = TRUE)}

  ukbox <- make_bbox(lon = c(min(lon), max(lon)), lat = c(max(lat), min(lat)), f = .1)
  uk <- get_map(location = ukbox, zoom = 10, maptype = "terrain", source = "google")
  dataframe <- data.frame(x = lon, y = lat, data = data)

  p <- ggmap(uk)
  p <- p + geom_tile(dataframe, mapping = aes(x = x, y = y, fill = data),
                    alpha = 1) + #
    labs(title=mainTitle, x="Longitude", y="Latitude") +
    theme(axis.text=element_text(size=18),
          axis.title=element_text(size=18),
          plot.title = element_text(size=20, hjust = 0.5),
          legend.text = element_text(size = 18),
          legend.title = element_text(size = 18),
          legend.key.size = unit(0.3, "in")) +
    scale_fill_viridis(name = legendTitle, limits = zlim) +
    xlim(xlim) + ylim(ylim)

  baseData <- map_data("world")
  p <- p + geom_path(data=baseData, aes(x=long, y=lat, group=group),
                    colour="black")

  return(p)}

p <- map.heatmap(lat = locs.pred[ , 2],
                 lon = locs.pred[ , 1],
                 data = pm2.5.pred,
                 mainTitle = "PM2.5 conc (ug/m3)")

```

! Bounding box given to Google - spatial extent only approximate.

i <<https://maps.googleapis.com/maps/api/staticmap?center=28.64,77.085&zoom=10&size=640x640&scale=2&m>

Scale for x is already present.

```
## Adding another scale for x, which will replace the existing scale.
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.
```

```
print(p)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_rect()').
```

```
## Warning: Removed 16 rows containing missing values or values outside the scale range
## ('geom_tile()').
```

```
## Warning: Removed 99338 rows containing missing values or values outside the scale range
## ('geom_path()').
```

